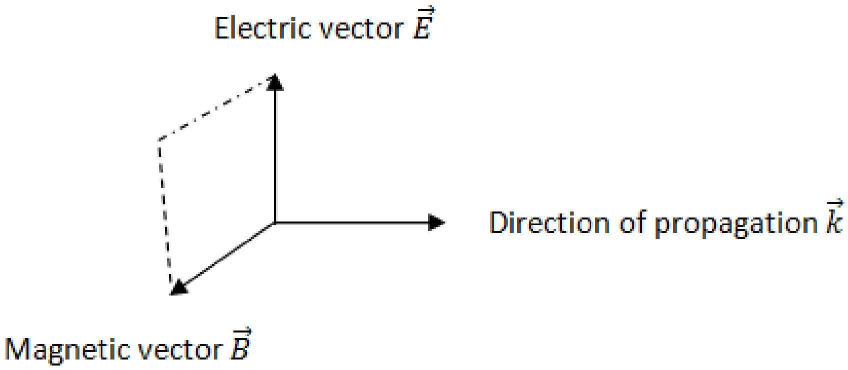
Q1. Two wires of equal length, one of copper and the other of manganin have the same resistance. Which wire is thicker?

Answer: We know that resistance of a wire is given by $R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}$

Since by nature resistivity ρ of manganin is much greater than that of copper, therefore to keep same resistance R, for same length of wire, the manganin wire (radius should be more) must be thicker.

Q2. What are the direction of electric and magnetic field vectors relative to each other and relative to the direction of propagation of electromagnetic waves?

Answer: In electromagnetic waves the electric and magnetic field vectors are perpendicular to each other and are perpendicular to the direction of propagation of wave.

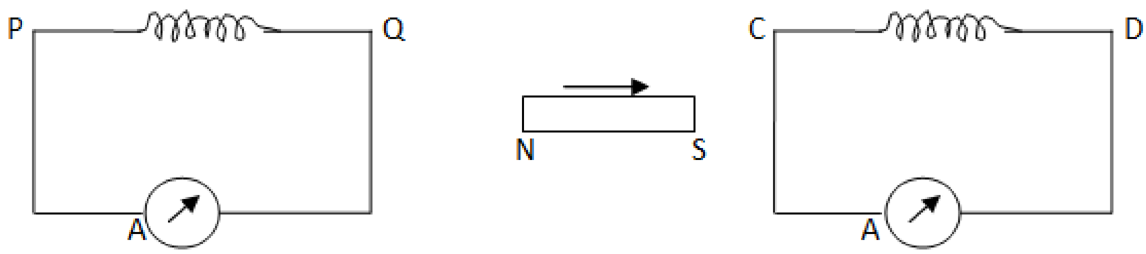


Q3. How does the angular separation between fringes in single-slit diffraction experiment change when the distance of separation between the slit and screen is doubled?

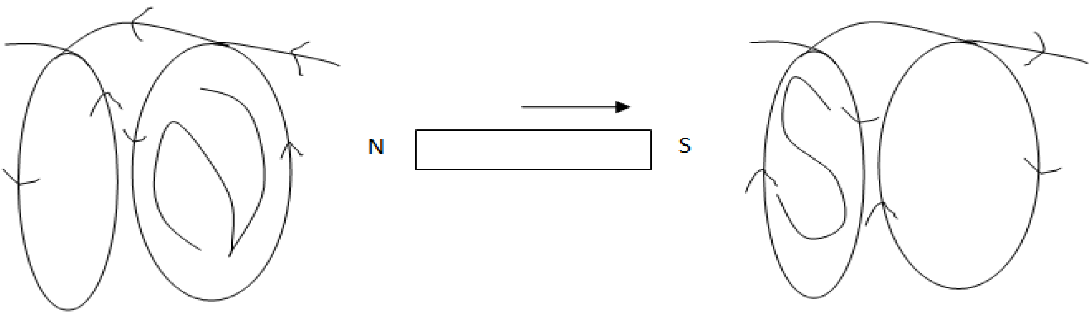
Answer: The angular width of central diffraction band $= \frac{2\lambda}{a} \propto \frac{1}{a}$, when slit width is doubled the width of central band becomes half of the initial value. As intensity of central maximum $\propto a^2$, if slit width is doubled, the intensity becomes four times.

Answer: The separation is not the function of distance beween the slit and the screen , it would not change

Q4. A bar magnet is moved in the direction indicated by the arrow between two coils PQ and CD. Predict the direction of induced current in each coil.



Answer: We know that as per Lenz's law direction of induced emf is such that it opposes the very cause due to which it is produced. So if South Pole approaches to CD then direction of current should be such that S pole produced in CD, so south will oppose approaching south (as shown), similarly if North Pole approaches PQ then direction of current in PQ is such that North Pole is produced in PQ.



Section of PQ Section of CD

Just if we write the letter S and N, direction of current is self explanatory.

Q5. For the same value of angle of incidence, the angles of refraction in three media A, B and C are 15^0, 25^0 and 35^0 respectively. In which medium would the velocity of light be minimum?

Answer: For Snell's law, $n = \frac{\sin i}{\sin r} = \frac{C}{v}$

For given i, $v \propto \sin r$

$\text{Sin}15^0 < \sin 25^0 < \sin 35^0$

Thus r is minimum in medium A, so velocity of light is minimum in medium A.

Q6. A proton and an electron have same kinetic energy. Which one has greater de-Broglie wavelength and why?

Answer: An electron has the larger wavelength.

Reason: de-Broglie wavelength in terms of kinetic energy is

$\lambda = \frac{h}{\sqrt{2mE_K}}$ or $\lambda \propto \frac{1}{\sqrt{m}}$ for the same kinetic energy. Mass of an electron is smaller than mass of proton therefore an electron has larger de-Broglie wavelength than a proton when both have the same kinetic energy.

Q7. Mention the two characteristic properties of the material suitable for making core of a transformer.

Answer:

1. High permeability: to reduce flux leakage, less hysteresis loss.
2. Soft magnetic property: to reduce loss of power as heat material must have soft magnetic property so that it can be magnetised and demagnetised with minimum heat loss.

Q8.A charge 'q' is placed at the centre of a cube of side *l*. What is the electric flux passing through each face of the cube?

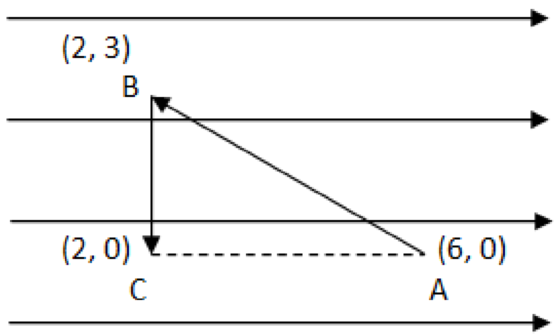
Answer:

According to Gauss's theorem, the total electric flux through the six faces of cube $= \frac{q}{\epsilon_0}$.

Therefore flux through each face = $\frac{q}{6\epsilon_0}$

Q9. A test charge 'q' is moved without acceleration from A to C along the path from A to B and then from B to C in electric field E as shown in the figure.

I. Calculate the potential difference between A and C.
II. At which point of the two is the electric potential more and why?



Answer: Distance between point A and C

AC = 6 – 2 = 4

$\therefore \mathrm{dr} = \mathrm{AC} = 4$

$\mathrm{E} = \frac{\mathrm{dV}}{\mathrm{dr}}$ Therefore Potential difference = $V_A - V_C$=dV= E.dr=E(6-2)=4E

Work done against field is stored as potential when it is against field work is more thus while moving from A to C work is done against the direction of electric field , i.e Potential at C will be more.

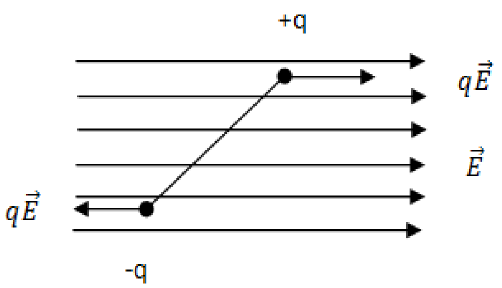
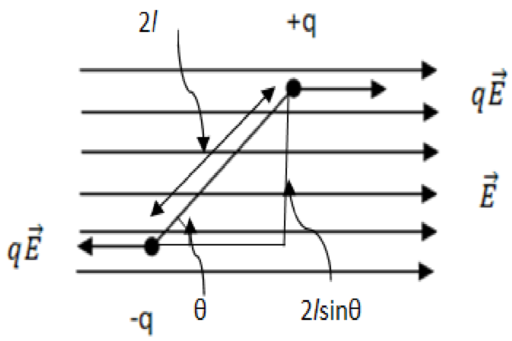
$\therefore V_C > V_A$

Q10. An electric dipole is held in a uniform electric field.

I. Show that the net force acting on it is zero.

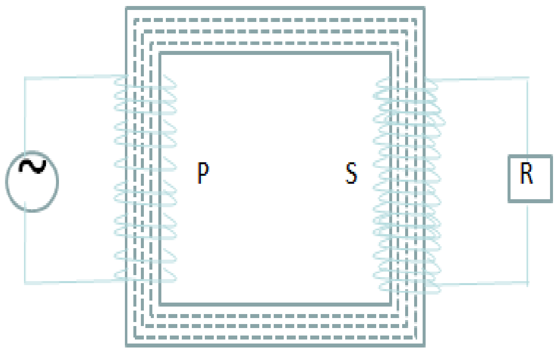
II. The dipole is aligned parallel to the field.
Find the work done in rotating it through the angle of 180^0.

Answer:

	I. Force on electric dipole kept in uniform electric field: An electric dipole is placed in uniform electric field ($\vec{E}$). The force acting on +q charge $q\vec{E}$ which is along the direction of electric field and force in –q charge is $q\vec{E}$ but this force is acting in opposite direction as shown in the figure. Since these two forces are equal in magnitude and opposite in direction hence net force is zero. However these two forces constitute couple.
	II. Moment of the force =force X perpendicular distance =qEx2*l*/sinθ=PEsinθ (P=qx2*l* = dipole moment) Work done to move by an angle dθ, dw = PEsinθdθ Therefore work done to rotate from θ=0 to θ=180^0 $W = \int dw = \int_0^{\pi} PESin\theta d\theta = PE[-cos\theta]_0^{\pi}$ $= -PE[cos\pi - cos0] = -PE[-1-1] = 2PE\ Joule$

Q11. State the underlying principle of transformer. How is the large scale transmission of electric energy over long distances done with the use of transformers?

Answer:



Principle: It is a device which converts high voltage, A.C. into low voltage A.C. and vice versa. It is based upon the principle of *mutual induction*. When alternating current passed through a coil (Primary), an induced e.m.f. is set up in the neighbouring coil(Secondary).

Construction: A transformer consists of two coils of many turns of insulated copper wire wound on a closed laminated iron core. One of the coils known as primary (P) is connected to A.C. supply. The other coil known as Secondary (S) is connected to the load.

Working: It consists of two coils known as Primary (P) and Secondary(S) having different no. of turns and are would on two opposite arms of a thick laminated iron core. The alternating emf to be transformed is connected across the primary coil. The varying current flowing through the primary coil produces varying magnetic lines of force. The magnetic ion core provides an easy path for the flow of lines of force hence the lines of force flowing through the core cuts the secondary coil and induces an emf across the secondary. An induced current flow through the secondary coil and thus a potential drop is obtained across the resistance which is known as output voltage.

Theory: Given

N_p and N_s are number of turns in the primary and secondary respectively

V_p and V_s are their respective voltages.

$\therefore V_p = -N_p \frac{d\phi}{dt}$ and $V_s = -N_s \frac{d\phi}{dt}$

Where, N_p and N_s are number of turns in the primary and secondary respectively and V_p and V_s are their respective voltages.

$\therefore \frac{V_s}{V_p} = \frac{N_s}{N_p}$ This ratio $\frac{N_s}{N_p}$ is called the turns ratio.

$In\ a\ step-up\ transformer:\ N_s > N_p,\ so\ V_s > V_p$.

$In\ a\ step-down\ transformer:\ \mathrm{N_s} < \mathrm{N_p}\ ,\ \ so\ \mathrm{V_s} < \mathrm{V_p}$

The larger scale transmission and distribution of electrical energy over long distances is done with the use of transformers. In order to reduce loss due to heat produced during transmission, for the power generation station using step up transformer high voltage low current transmitted(Heat=i^2Rt) since current is low hence less heat is produced hence less loss. Near consumers locality area sub-station it is stepped down low voltage but high current. It is further stepped-down at distributing sub-stations and utility poles before a power supply of 240 V reaches our homes.

Q12. A capacitor of capacitance 'C' is being charged by connecting it across a *dc* source along with an ammeter. Will the ammeter show a momentary deflection during the process of charging? If so, how would you explain this momentary deflection and the resulting continuity of current in the circuit? Write the expression for the current inside the capacitor.

Answer: Yes, during the process of charging ammeter will show momentary deflection.

When the Capacitor is connected, initially charge in capacitor is zero, as the charge flows potential will raises. The flow of charge continues till the potential difference across the capacitor becomes equal to the emf of the source. Thus as long as charge flows it gives current in the circuit and ammeter will show deflection. When the capacitor gets fully charged there will not be any difference in potential so at steady state charge flow will be negligible (almost zero), deflection in ammeter will not be observed.

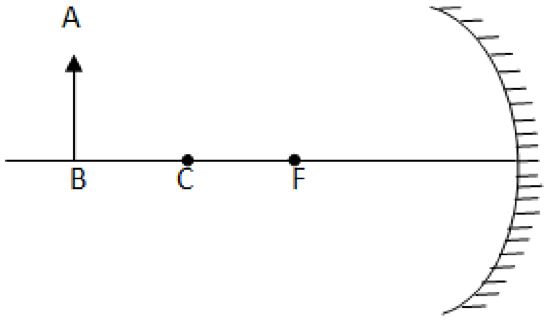
$$e.m.f\ equataion\ \frac{Q}{c} + iR = E, Solving\ we\ get\ Q = Q_0\left(1 - e^{t/_{RC}}\right) where\ Q_0 = EC$$

R= Resistance, E= emf of dc source, C=Capacitance this gives the growth of charge.

Since $i = \frac{dQ}{dt} = i = \frac{Q_0}{RC}e^{-t/_{RC}} = i_0 e^{-t/_{RC}}\ where\ i_0 = \frac{Q_0}{RC}$.

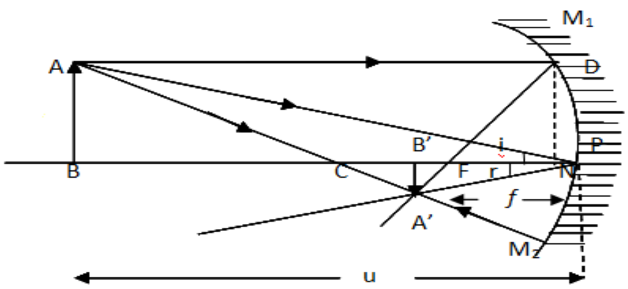
Expression for current $i = i_0 e^{-t/_{RC}}$

Q13. An object AB is kept in front of a concave mirror as shown in the figure.



a) Complete the ray diagram showing the image formation of the object.
b) How will the position and intensity of the image be affected if the lower half of the mirror's reflecting surface is painted black?

Answer: (a) Ray Diagram:

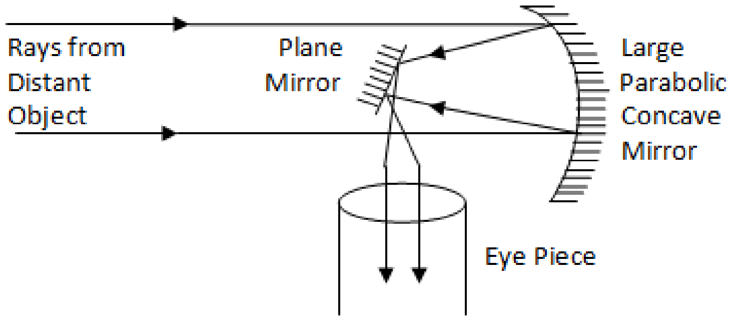


(b) On painting lower half black the ray AC will be absorbed and hence the intensity of the image will decrease, image will be formed in the same position.

Q14. Draw a labelled ray diagram of a reflective type telescope. Write its two advantages over refracting type telescope.

Answer:

Ray diagram:



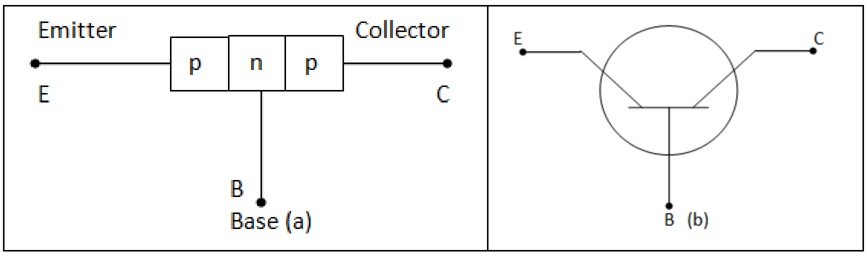
Two advantages:

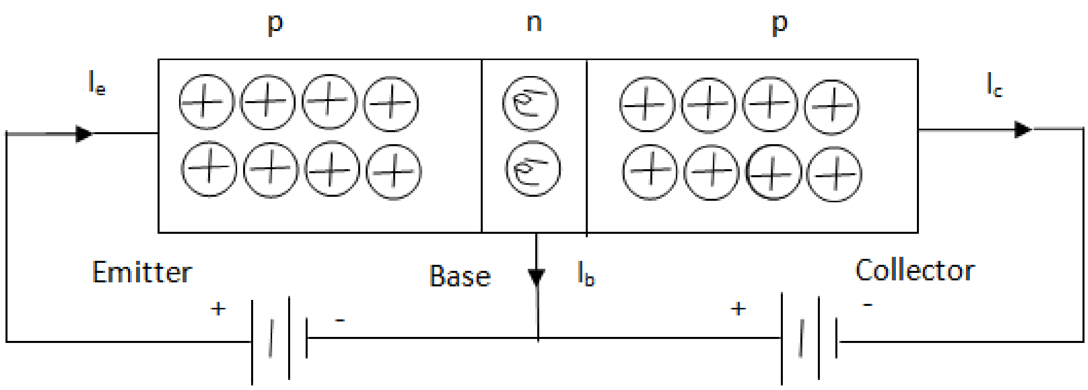
A. As a reflecting telescope has mirror objective, the brighter image formed is free from chromatic aberration.

B. It is comparatively cheaper than refracting type telescope.

Q15. Describe briefly with the help of a circuit diagram, how the flow of current carriers in a p-n-p transistor is regulated with emitter-base junction forward biased and base-collector junction reverse biased.

Answer: p-n-p transistor: It is a single crystal containing two p-n junctions such that there is a very thin central layer of n-type semiconductor enclosed on either side by p-type semiconductor. The central layer is called the base.

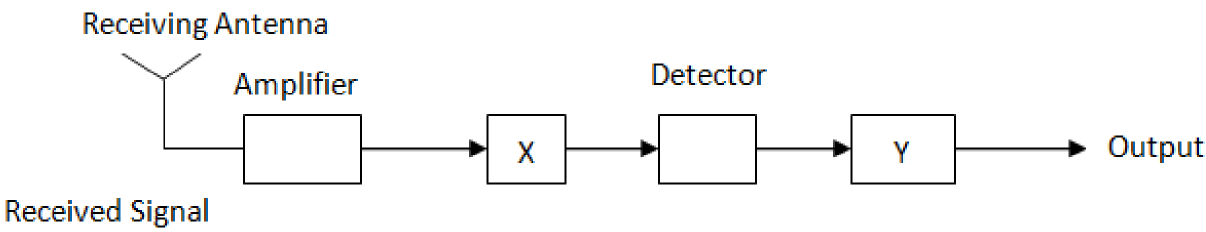




As shown above when p-n junction (emitter –base) junction is forward biased, p connective to positive terminal and n connected to negative terminal of battery. Holes from p moves towards n as emitter base junction is conducting through thin width layer. Some holes and electrons get neutralized and depletion layer is formed. A small current I_b flows, most of the holes moves towards collector giving collector current I_c. In the circuit above the base collector junction is reversed biased.Thus the current is from emitter to base.

Emitter current $I_e = I_b + I_c$

Q16. In the given block diagram of receiver, identify the boxes labelled as X and Y and write their function.



Answer: X is Intermediate Frequency (IF Stage) stage; here the received signal for antenna is converted to lower frequency signal call Intermediate Frequency.

Y is amplifier, which amplifies signal to produce desired output.

Q17. A light bulb is rated 100 W for 220 V ac supply of 50 Hz. Calculate

I. The resistance of the bulb;
II. The *rms* current through the bulb.

Answer:

I. The resistance of the bulb,

$$\mathrm{R} = \frac{\mathrm{E}^2}{\mathrm{P}} = \frac{(220)^2}{100} = \frac{48400}{100} = 484\ \Omega$$

ii. $I_{rms} = \frac{E}{R} = \frac{220}{484}$=0.45 A

Or, An alternating voltage given by $\mathrm{V} = 140 \sin 314\ \mathrm{t}$ is connected across a pure resistor of $50\ \Omega$. Find

I. The frequency of the source.
II. The rms current throughout the resistor.

Answer: Given $\mathrm{V} = 140 \sin 314\ \mathrm{t}$, comparing with $e=e_o \sin\omega t$, here ω=314Hz and e_o =140 volt

I. Frequency source $\mathrm{f} = \frac{\omega}{2\pi} = \frac{314}{2\times 3.14} = \frac{314\times 100}{314\times 2} = 50$ Hz

II. $\mathrm{e}_{rms} = \frac{\mathrm{e}_0}{\sqrt{2}} = \frac{140}{\sqrt{2}} = \frac{140}{1.41} = 99.29$ Hz.

rms value of current $i_{rms} = \frac{\mathrm{e}_{rms}}{\mathrm{R}} = \frac{99.29}{50} = 1.98\ \mathrm{A}$

Q18.A circular coil of N turns and radius R carries a current I. It is unwound and rewound to make another coil of radius R/2, current I remaining the same. Calculate the ratio of the magnetic moments of the new coil and the original coil.

Answer: We know that Magnetic Moment of coil M = NIA

Where M = Magnetic Moment of coil

N = Number of turns

I = current through the coil

A= Area of the coil.

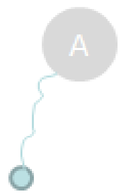
Let M' be the magnetic moment when radius changed to R/2 , therefore

Since length is same L = L', $2\pi RN = 2\pi R'N'$ but $R' = \frac{R}{2}$ therefore $2\pi RN = \frac{2\pi R}{2}N'$so $N' = 2N$

So $M' = N'IA' = 2NI\pi\left(\frac{R}{2}\right)^2 = \frac{NIR^2}{2} = \frac{M}{2}$ so $\frac{M'}{M} = \frac{1}{2}$

Q19. Deduce the expression for the electrostatic energy stored in a capacitor of capacitance 'C' and having charge 'Q'. How will the (I) energy stored and (II) the electric field inside capacitor be affected when it is completely filled with a dielectric material of dielectric constant 'K'?

Answer: Expression for electrostatic energy stored in a capacitor

 Source of Potential V	Given C = Capacitance of the body. V= Potential of the source Q= The maximum charge stored in the body when its potential rises to V Let during charging q be the charge stored in the body at any instant t and v be the potential of the body at that instant. $q = cv$ $v = \frac{q}{c} \rightarrow (1)$

The charge q which is already in the body will repel the like charge dQ as it comes from the Source hence work is to be done in storing the charge dQ

Hence total work done in storing the charge Q can be obtained by integrating equation(2)

$$W = \int dw = \int_0^Q \frac{q}{C} dQ = \frac{1}{2}\frac{Q^2}{C} \rightarrow (3)$$

This work done is stored in the charged body in the form of electrostatic potential energy.

$$\therefore E = \frac{1}{2}\frac{Q^2}{C} \rightarrow (4)$$

$$Q = CV$$

$$E = \frac{1}{2}\frac{C^2V^2}{C} = \frac{1}{2}CV^2$$

$$E = \frac{1}{2}C(Q/C)^2 = \frac{Q^2}{2C} \rightarrow (5)$$

Electric field inside a capacitor:

We know that capacitance of capacity $C = \frac{\varepsilon A}{d}$

When there is free space or air that is no dielectric between plates of capacitor then $C = \frac{\varepsilon_0 A}{d}$

When medium of dielectric constant K is introduced permittivity changes i.e. $\varepsilon = k\varepsilon_0$

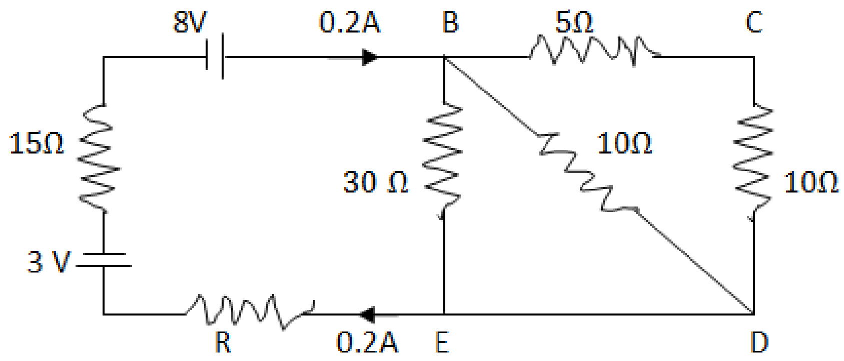
Therefore $C' = \frac{k\varepsilon_0 A}{d} = kC$

If E and E' be the energy stored with free space and dielectric (K) then

$$E = \frac{Q^2}{2C}, E' = \frac{Q^2}{2C'} = \frac{Q^2}{2KC} = \frac{1}{K}E$$

Thus with the introduction of dielectric constant the energy of capacitor becomes 1/K times initial energy.

Q20. Calculate the value of the resistance R in the circuit shown in the figure so that the current in the circuit is 0.2 A. What would be the potential difference between points B and E?

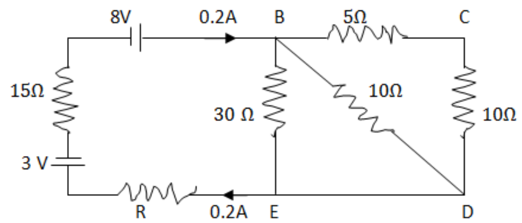


Answer: Looking at the circuit given resistance in BC arm in series with resistance in CD arm so Resistance between B to D = 5 + 10 =15 Ω . Also this 15 Ω is in parallel to resistance in arm BD and BE

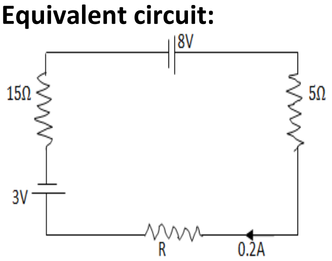
Total resistance of the loop BCDEB, $\frac{1}{R} = \frac{1}{15} + \frac{1}{10} + \frac{1}{30}$

$\frac{1}{R\prime} = \frac{2+3+1}{30} = \frac{6}{30} = \frac{1}{5} \quad \therefore R' = 5\,\Omega$

Actual Circuit



Equivalent circuit:



Net potential = 8 – 3 = 5V

$\therefore I = \frac{V}{R}$

$0.2 = \frac{5}{5+R+15}$

$0.2 = \frac{5}{20+R}$

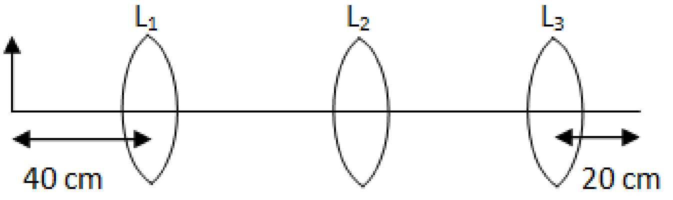
$\Rightarrow 0.2(20 + R) = 5$

$\Rightarrow R = \frac{1}{0.2} = 5\,\Omega$

Now Potential difference between B and E

V_{BE} = 0.2x 5 =1 Volt

Q21. You are given three lenses L_1, L_2 and L_3, each of focal length 20 cm. An object is kept at 40 cm in front of L_1, as shown. The final real image is formed at the focus 'I' of L_3. Find the separations between L_1, L_2 and L_3.

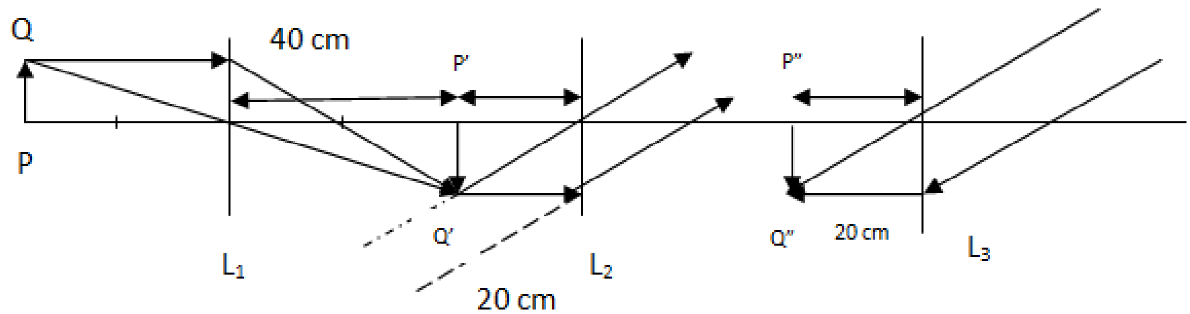


Answer: Considering the first lens L_1 , focal length=20 cm, u=40 cm, it means object PQ is kept at 2f from convex lens L_1 hence image P'Q' will be formed at 2f i.e 40 cm on the other side of L1.

Now let us take it in reverse way, final image P"Q" is formed by at focus of L_3 (at a distance 20 cm), this is possible if object distance for L_3 is at infinity (object of L_3 not shown in ray diagram as it is at infinite distance and rays falling on L_3 is parallel to each other).

L_2 produced image is actually object for L_3, so L_2 must produce image at infinity then only L_3 can form final image at its focus. This is possible if object (P'Q') distance for L_2 is equal to its focal length.

Case I: If the distance of separation is more than 40 then one possibility is object distance for L_2 is equal to its focal length 20 cm therefore separation between L_1 and L_2 is 40+20=60 cm.



Case II: if separation is not more than 40 cm then P'Q' will lie on the other side of L_2, above ray diagram cannot be justified.

Thus separation between lenses cannot be concluded with certainty.

Q22. Define the terms (I) cut-off voltage and (II) threshold frequency in relation to phenomenon of photoelectric effect.

Using Einstein's photo electric equation show how the cut-off voltage and threshold frequency for a given photosensitive material can be determined with the help of suitable photograph.

Answer: (I) Cut-off voltage: When the collector is kept at negative potential with respect to the emitter current (i) decrease rapidly as Potential (V) become more and more negative and current (i) becomes zero at a particular value of negative potential. This negative potential at which photo electric current becomes zero is known as stopping potential (V_s) or Cut-off voltage.

(II) Threshold frequency: For every emitting electrode there is a minimum value of frequency of the incident light below which no photoelectric current is emitted and that frequency is known as threshold frequency.

Einstein's Photo electric equation:

Let ν = frequency of incident light

E = Energy of the incident light = $h\nu$

h = Plank's constant = 6.625×10^{-34} Joule Sec

$\mathrm{E_0}$ = Work function of the emitting electrode

For emission of electron the necessary condition $\mathrm{E} >> \mathrm{E_0}$

$$\mathrm{E_0} = h\nu_0$$

$$\therefore h\nu \geq h\nu_0$$

$$\nu \geq \nu_0$$

Thus for emission of photo electron the frequency of incident light ν must be greater than or equal to ν_0 hence ν_0 is the thereshold frequency.

Let $\mathrm{E} > \mathrm{E_0}$

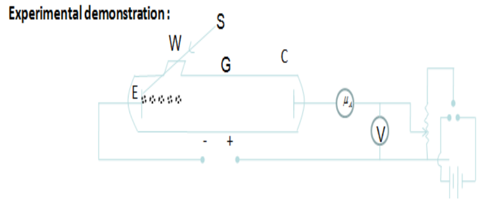
$E - E_0$ = The maximum kinetic energy of the phto electron $= \frac{1}{2}mv_{max}^2$

$$\text{or } h\nu - h\nu_0 = \frac{1}{2}mv_{max}^2$$

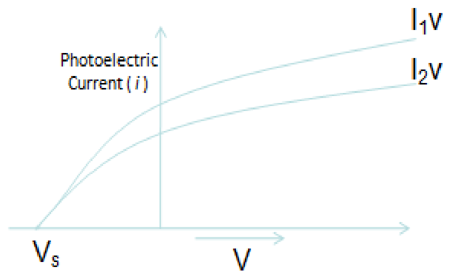
$$h\nu - h\nu_0 = eV_s$$

$$\nu - \nu_0 = \frac{e}{h}V_s$$

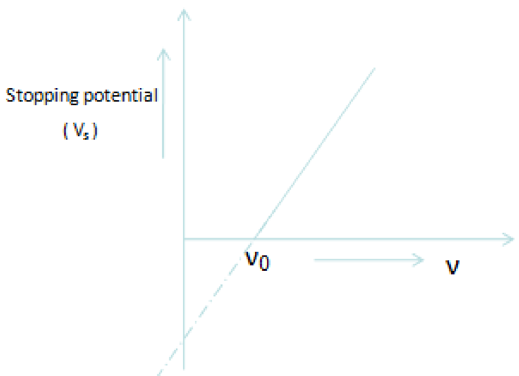
$$\nu = \nu_0 + \frac{e}{h}V_s$$



Graph for finding Stopping Potential:



Graph of finding Threshold frequency



G is a glass discharge tube having a window W of quartz which allow the ultraviolet light to pass through. E is the emitting surface and C is the collecting electrode. A potential difference known as accelerating potential difference is applied between the two electrodes by using a potential divider arrangement and commutator and he applied potential difference is measured by a voltmeter. The photo electric current flowing in the circuit is measured by micro ammeter.

Photo electric equation is $\nu = \nu_0 + \frac{e}{h} V_s$

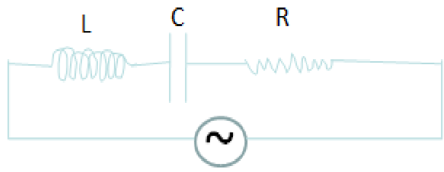
1. Different voltages applied plotted along X-axis and photo electric current found plotted along Y axis. Voltage for which no current found in circuit is cut-off voltage (Graph shows Vs data point)

2. Now keeping intensity of incident light same only Frequency is varied from the plotted data point graph shown threshold frequency can be found.

Q23. A series LCR circuit is connected to an ac source. Using the phasor diagram, derive the expression for the impedance of the circuit. Plot a graph to show the variation of current with frequency of the source, explaining the nature of its variation.

Answer:

L-C-R circuit :



We know that in capacitor (C) current leads emf by π/2 , in inductance current lags emf by π/2 (L) and in resistance (R) current is in phase with emf.

If V_L, V_c, V_R and V represents the voltage across the inductor, capacitor, resistance and source.

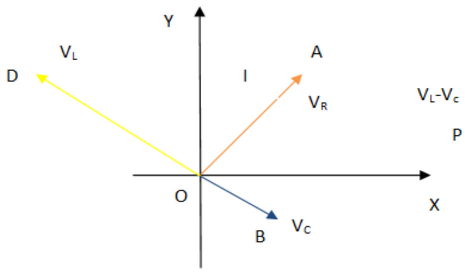


Fig-1

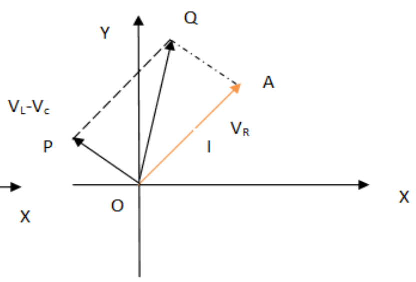


Fig-2

For the above vector diagram, Let the direction of current be along $\overrightarrow{OA}$, therefore

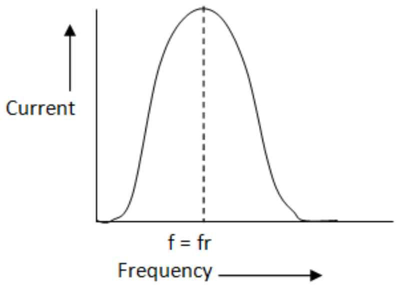
V_R= IR can be represented along $\overrightarrow{OA}$.
V_L=IX_L=IωL , since it lags current by π/2 , represented along $\overrightarrow{OD}$.
V_c=IXc=I/ωC, since current leads by π/2, represented along $\overrightarrow{OB}$.
Therefore V_c and V_L are 180^0 apart, resultant obtained by subtraction= V_L- V_C=$\overrightarrow{OP}$ (shown in fig-2)

Now from ΔOQP: $OQ^2 = OP^2 + PQ^2$, $V^2= V_R^2+(V_L -V_c)^2= I^2[R^2 +(\omega L- 1/\omega C)^2]$ Or $\frac{V}{I} = Impedance(Z)$

The impedance of the circuit

$$z=\sqrt{R^2+(X_L-X_C)^2}=\sqrt{R^2+\left(\omega L-\tfrac{1}{\omega c}\right)^2}$$

The given graph showing variation of current with the frequency of the applied voltage.



As the frequency increased initially current increases, for a particular value of frequency current reaches to its peak value (maximum value) in this instance Impedance offered by LCR is minimum, this frequency is known as Resonant Frequency ω_r. When frequency of AC in increased further current decreases. The rapidity by which current increases or decreases with frequency is referred as sharpness, more is the sharpness less will be the inverted bell graph shown width. It is called acceptor circuit.

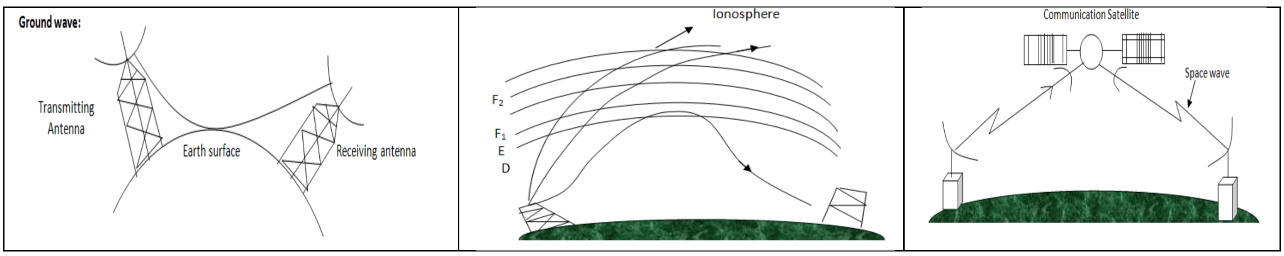
The impedance of the circuit

$$z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Z will be minimum where when $\omega L = \frac{1}{\omega C}$ $or\ \omega^2 = \frac{1}{LC}$ $this\ \omega\ say\ \omega_r\ \ therefore\ \omega_r = \frac{1}{\sqrt{LC}}$

Q24. Mention three different modes of propagation used in communication system. Explain with the help of diagram how long distance communication can be achieved by ionospheric reflection of radio waves.

Answer: schematic diagram showing the (I) ground wave (II) sky wave and (III) space wave



The ionosphere is a region of the upper atmosphere where there are large concentrations of free ions and electrons. Ionosphere contains ions, the free electrons that affect the radio waves and radio communications. The ionosphere affects signals on the short wave radio bands where it "reflects" signals, which helps these radio communications signals to be heard over vast distances. Radio stations have long used the properties of the ionosphere to enable them to provide worldwide radio communications coverage. Although today, satellites are widely used, HF radio communications using the ionosphere still plays a major role in providing worldwide radio coverage.

The different layers of ionosphere contributing to the wave propagation are

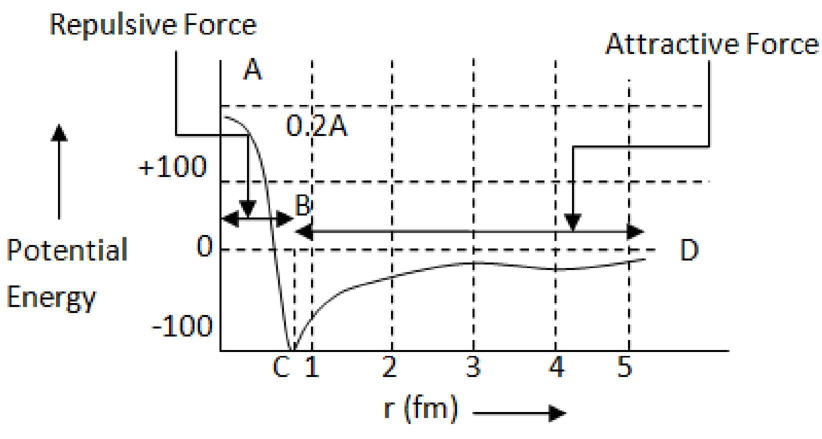
D- Region:60-90 KM, It mainly has the affect of absorbing or attenuating radio communications signals particularly in the LF and MF portions of the radio spectrum.

E-Region: Above D-Region, 100 and 125 kilometres. Instead of attenuating radio communications signals this layer chiefly refracts them, often to a degree where they are returned to earth.

F-Region: It is subdivided into F1 and F2, F1 at height 300 KM & F2 layer above it at around 400 kilometres. The combined F layer acts as a "reflector" of signals in the HF portion of the radio spectrum enabling worldwide radio communications to be established. It is the main region associated with HF signal propagation.

Q25. Draw a plot of potential energy of a pair of nucleons as a function of their separations. Mark the regions where the nuclear force is (I) attractive and (II) repulsive. Write any two characteristic features of nuclear forces.

Answer:



Potential energy is plotted along Y –axis, distance of separation between nucleons r(fm) plotted along X-axis, in the above graph segment AB represents repulsive force and BCD segment represents attractive force.
Two characteristic features of Nuclear Forces

i. Nuclear forces are stronger but short ranged.
ii. Nuclear forces are charge-independent.

Q26. In a Geiger – Marsden experiment, calculate the distance of closet approach to the nucleus of Z = 80, when an $\alpha -$ particle of 8 MeV energy impinges on it before it comes momentarily to rest and reverses its direction.

How will the distance of closet approach be affected when the kinetic energy of the $\alpha -$ particle is doubled?

Answer: At the closest distance the Kinetic Energy = Electric Potential Energy

Given Kinetic Energy (KE) = 8 MeV $= 8 \times 1.6 \times 10^{-13}$J.

Potential energy $= \frac{1}{4\pi\epsilon_0}\frac{(\mathrm{Ze})\mathrm{e}}{\mathrm{r}} = 9 \times 10^9 \times \frac{80\times(1.6\times10^{-19})^2}{\mathrm{r}}$

Therefore $8 \times 1.6 \times 10^{-13} = 9 \times 10^9 \times \frac{80\times(1.6\times10^{-19})^2}{\mathrm{r}}$

At closest approach: $9 \times 10^9 \times \frac{80\times(1.6\times10^{-19})^2}{\mathrm{r}} = 8 \times 1.6 \times 10^{-13}$

$$\mathrm{r} = 9 \times 10^9 \times \frac{80 \times (1.6 \times 10^{-19})^2}{8 \times 1.6 \times 10^{-13}} = \frac{9 \times 1.6 \times\ 10^{9-38+13+1}}{1} = 14.4 \times 10^{-15}\mathrm{m}$$

When KE is doubled it becomes = 16 MeV, then Kinetic Energy = Electric Potential Energy

$16 \times 1.6 \times 10^{-13}$=$9 \times 10^9 \times \frac{80\times(1.6\times10^{-19})^2}{\mathrm{r}_1}$ here distance of closest approach $\frac{14.4\times10^{-15}}{2}$

Thus the distance of closed approach becomes half.

Q27. Define relaxation time of the free electrons drifting in conductor. How is it related to the drift velocity of free electrons? Use this relation to deduce the expression for the electrical resistivity of the material.

Answer: Relaxation time **-** The time of free travel of electrons between successive collisions is called the relaxation time. It is denoted by τ.

Relation between relaxation time (τ) and drift velocity(v_d):

Let v_d= drift velocity is the average velocity that a particle, such as an electron, attains due to an electric field.

m= mass of electron, e= charge of electron, E = Electric field then

$$\tau = \frac{v_d m}{eE} \rightarrow (1)$$

When temperature increases, the thermal speed of electrons increases; so collisions of electrons occur more frequently. As a result the relaxation time of electrons decreases.

Expression for electrical Resistivity:

The formula for evaluating the drift velocity of charge carriers in a material of constant cross-sectional area is given by

$$\overrightarrow{v_d} = \frac{I}{neA} \ or \ I = \overrightarrow{v_d}neA \rightarrow (2)$$

Where I = current flowing through the material,

n = charge-carrier density

A = area of cross-section of the material

Multiplying both sides of equation (2) by Δt, we get

$$I\Delta t = \overrightarrow{v_d}neA\Delta t$$

$\frac{I\Delta t}{A\Delta t} = \overrightarrow{v_d}ne$ Putting the value of $\overrightarrow{v_d}$ from equation (1)

$$\vec{J} = \frac{\tau e\vec{E}}{m}ne$$

$$\vec{J} = \frac{n\tau e^2 \vec{E}}{m} \rightarrow (3)$$

Because current is proportional to drift velocity, which in a resistive material is, in turn, proportional to the magnitude of an external electric field.

We know that resistivity (ρ) is defined as $\rho = \frac{E}{J}$ $from$ (3)

$$\rho = \frac{m}{n\tau e^2} \rightarrow (4)$$

Equation (4) gives the relation between resistivity and relaxation time.

Q28.

a) In Young's double slit experiment, derive the condition for (I) constructive interference and (II) destructive interference at a point on the screen.

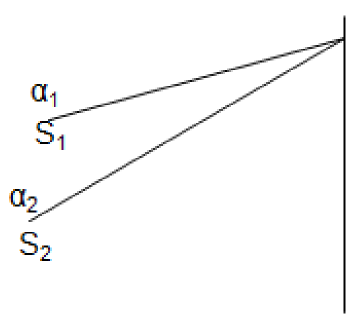
b) A beam of light consisting of two wavelengths, 800 nm and 600 nm is used to obtain the interference fringes in a Young's double slit experiment on a screen placed 1.4 m away. If the two slits are separated by 0.28 mm, calculate the least distance from the central bright maximum where the bright fringes of the two wavelengths coincide.

Answer: (a) Let d = distance of separation between two slits.

D = Perpendicular distance of the screen from the slits

a & R = amplitude of the waves emitted from S_1 and S_2 respectively.

λ = wave length of light ($\omega = 2\pi\nu = 2\pi c/\lambda$)



$$\beta_1 = \alpha_1 + \frac{2\pi}{\lambda} x_1$$

$$\beta_2 = \alpha_2 + \frac{2\pi}{\lambda} x_2$$

$$\phi = \beta_2 - \beta_1 = \alpha_2 + \frac{2\pi}{\lambda} x_2 - \left(\alpha_1 + \frac{2\pi}{\lambda} x_1\right)$$

$$\phi = (\alpha_2 - \alpha_1) + \frac{2\pi}{\lambda}(x_2 - x_1)$$

Let y_1 and y_2 be the displacement at any point P on the screen at an instant of time t of the waves emitted from S_1 and S_2 respectively. Using displacement equation:

$$y_1 = a \sin \omega t \rightarrow (1)$$
$$y_2 = a \sin (\omega t \pm \phi) \rightarrow (2)$$
$$\phi = (\alpha_2 - \alpha_1) + \frac{2\pi}{\lambda}(x_2 - x_1) \rightarrow (3)$$
$$y = y_1 + y_2 = a \sin \omega t + a \sin (\omega t \pm \phi)$$
$$y = a \sin \omega t + b \sin \omega t . \cos \phi \pm b \cos \omega t . \sin \phi$$
$$y = (a + b \cos \phi) \sin \omega t \pm b \cos \omega t . \sin \phi \rightarrow (4)$$
$$Putting \;\; \text{bsin } \phi = \text{csin } \delta \rightarrow (5)$$
$$\text{a + bcos } \phi = \text{ccos } \delta \rightarrow (6)$$
$$\text{Putting equation (5) and (6) in (4)}$$
$$\text{y} = \text{csin } \omega\text{t.cos } \delta \pm \text{c cos } \omega\text{t.sin } \delta$$
$$\text{y} = \text{csin } (\omega\text{t} \pm \delta) \rightarrow (7)$$

From equation (7) we find that c is the amplitude of the resultant wave due to the superposition. Squaring and adding equation (5) & (6)

$$c^2 = b^2 \sin^2 \phi + a^2 + b^2 \cos^2 \phi + 2ab \cos\phi$$
$$c^2 = a^2 + b^2 + 2ab\cos\phi \rightarrow (8)$$

I = Resultant intensity at P due to the superposition of the two waves

$$I \propto (amplitude)^2$$
$$I \propto c^2$$
$$I = \text{constant.c}^2$$
$$I = \text{constant.}\left[a^2 + b^2 + 2ab\cos\phi\right] \rightarrow (9)$$

(I)Condition for constructive interference:

$$\text{From equation (9) we find I is maximum when } \cos\phi \text{ is maximum.}$$
$$I = \text{constant}\left[a^2 + b^2 + 2ab\cos\phi\right]$$
$$\cos\phi = 1 = \cos 0 = \cos 2\pi = \cos 4\pi = \cos 6\pi = \ldots$$
$$\phi = 2n\pi \rightarrow (10) \text{ where n} = 0,1,2,3,4,\ldots$$
$$\text{or } \frac{2\pi}{\lambda}\Delta = 2n\pi \rightarrow (11)$$
$$\Delta = \frac{2n\pi}{2\pi}\lambda = 2n\frac{\lambda}{2} \rightarrow (12)$$
$$\text{where n} = 0,1,2,3,\ldots \text{an integer}$$
$$\Delta = 0, \lambda, 2\lambda, 3\lambda, 4\lambda \ldots\ldots$$

Path difference should be even multiple of λ/2.

(II) **Condition of minimum intensity (Destructive interference):** From equation (9) intensity is minimum when cos∅ is minimum i.e. cos∅= -1=cosπ=cos2π=cos3π,...

i.e. ∅ = π, 3π, 5π, 7π, ...

∅ = (2n+1) π → (13)

where n=0,1,2,3,4,..

$$\therefore \frac{2\pi}{\lambda}\Delta = (2n+1)\pi$$

$$\text{or } \Delta = (2n+1)\frac{\lambda}{2} \rightarrow (14)$$

Where n = 0,1,2,3,... .. an integer

$$\text{i.e. } \Delta = 1.\frac{\lambda}{2},\ 3\frac{\lambda}{2},\ 5\frac{\lambda}{2},\ 7\frac{\lambda}{2},\ldots\ldots\ldots$$

Thus for destructive interference path difference should be odd multiple of λ/2.

b) Position of first bright fringe,

Where, $\lambda_A = 800\text{ nm} = 800\times10^{-9}\text{m} = 8\times10^{-7}\text{m}$ and $D = 1.4\text{ m},\ d = 0.28\text{ mm} = 0.28\times10^{-3}\text{m}$

$X_A = n_1\lambda_A\frac{D}{d}$

For wavelength, $\lambda_B = 600\text{ nm} = 600\times10^{-9}\text{m} = 6\times10^{-7}\text{m}$

Position of first bright fringe is

$X_B = n_2\lambda_B\frac{D}{d}$

Since fringes coincides hence X_A=X_B

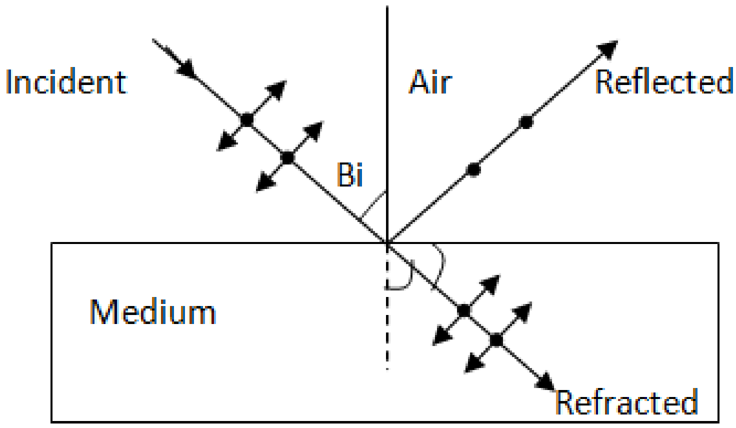
Or $n_1\lambda_A\frac{D}{d} = n_2\lambda_B\frac{D}{d}, \frac{n_1}{n_2} = \frac{\lambda_B}{\lambda_A} = \frac{600}{800} = \frac{3}{4}$, this is possible if n_1=3 and n_2=4 therefore common distance=

$X_A = n_1\lambda_A\frac{D}{d}$=$3\times8\times10^{-7}\times\frac{1.4}{0.28\times10^{-3}} = 120\times10^{-4} = 0.012\text{m} = 12\text{mm}$

OR

- **(a)** How does an unpolarized light light incident on a Polaroid get polarized? Describe briefly,with the help of a necessary diagram, the polarization of light by reflection from a transparent medium.
- **(b)** Two Polaroid's 'A' and 'B' are kept in crossed position. How should a third Polaroid 'C' be placed between them so that the intensity of polarized light transmitted by Polaroid B reduces to 1/8th of the intensity of unpolarized light incident on A?

i. **Answer:**



The independent light waves whose planes of vibrations are randomly oriented about the direction of propagation are said to be unpolarized light. When unpolarized light is incident on the boundary between two transparent media, the reflected light is polarized with electric vector perpendicular to the plane of incidence. Whenever unpolarized light is incident from air to a transparent medium at an angle of incidence equal to polarizing angle, the reflected light gets polarized as shown in the diagram above. Thus Light can be polarized by reflecting it from a transparent medium. The extent of polarization depend on the angle of incidence, called Brewster's angle.

Let μ= refractive index of the transparent medium

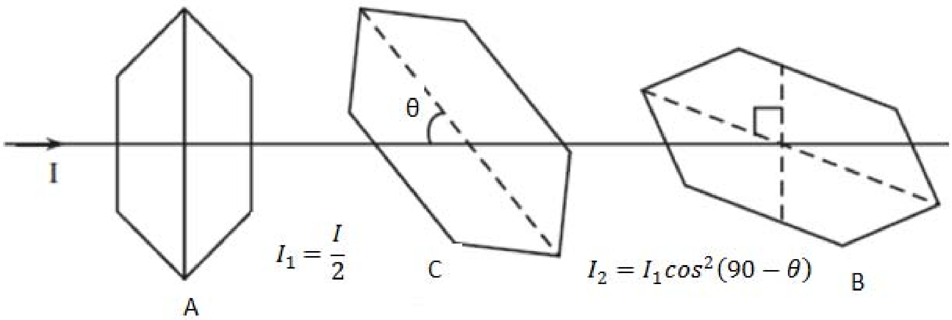
Bi= Brewster angle then

$\mu = tanB_i \ also \ \alpha + \delta = 90^0$

The refracted and reflected rays make a right angle with each other

(b) Given: A and B are kept at crossed position, intensity of light passing through A is half of intensity falling on A. Final output intensity from C is 1/8th of intensity falling on A.

Let θ = angle of inclination of C with the axis of A then if I, I_1,I_2 and I_3 be the intensity of light falling on A, passing though A, passing through B and passing through C respectively then



$$I_3 = \frac{I}{8} = I_2 cos^2\theta$$
$$= I_1 cos^2(90-\theta)cos^2\theta$$
$$= \frac{I}{2} sin^2\theta\, cos^2\theta$$

As output intensity calculated above

$$I_3 = \frac{I}{8} = I_2 cos^2\theta = I_1 cos^2(90-\theta)cos^2\theta = \frac{I}{2} sin^2\theta cos^2\theta$$

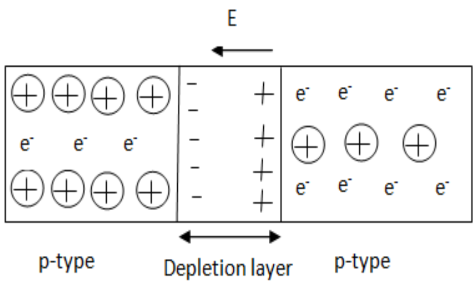
$$\frac{I}{8} = \frac{I}{2} sin^2\theta cos^2\theta \ or \ 1 = 4sin^2\theta cos^2\theta \ \ or \ 1 = sin2\theta \ or \ 2\theta = 90 \ or \ \theta = 45^0$$

Thus Polaroid C should be kept at 45^0 with the direction of axis of A.

Q29 (a) Describe briefly, with the help of a diagram, the role of the two important processes involved in the formation of a p-n junction.

(b) Name the device which is used as a voltage regulator. Draw the necessary circuit diagram and explain its working.

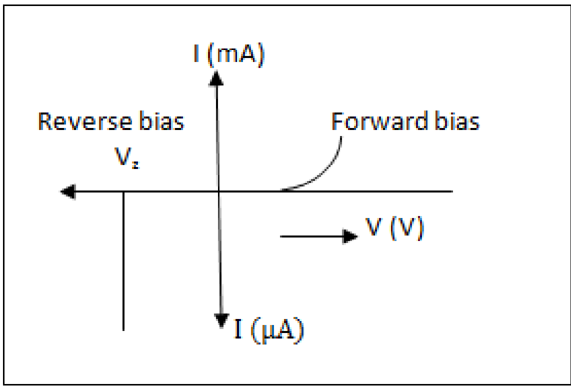
Answer:

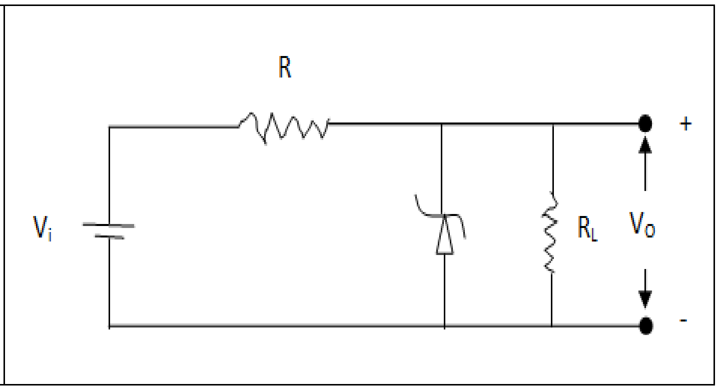
	As shown in p side we have higher concentration of holes majority charge carriers and in n side we have higher concentration of electrons. Therefore diffusion of holes that is electrons to the right of the junction may fill up the vacancies to the left of the junction. Similarly because of concentration difference conduction electrons diffuse from right to left. This makes left half positively charged and right half negatively charged. Electric field will be created from right to left, holes will be pushed to the left half and conduction electrons pushed towards right half. This region where no charge carrier remains is depletion region.

Diffusion current: Because of concentration difference holes having higher kinetic energy can cross the depletion region by overcoming the force due to electric field. Similarly diffusion electrons from right to left having higher kinetic energy can cross the junction opposed by field. The electric potential of the n-side is higher than that of the p-side, there is potential barrier at the junction which allows only a small amount of diffusion. This diffusion results electric current from p side to n side called diffusion current.

Drift current: Thermal collision results breaking of covalent bond and electrons jumps to the conduction band, electron hole pair gets created. Also conduction electron fills up vacant bond and electron-hole pair is destroyed. These processes continue in every part of the material. However if an electron-hole pair is created in the depletion region, electron is pushed by electric field towards n-side and holes towards P-side. Thus there is a regular flow of electrons towards n-side and holes towards p-side, this generates current from n-side to p-side called drift current.

(b) Zener diode is used as voltage regulator.



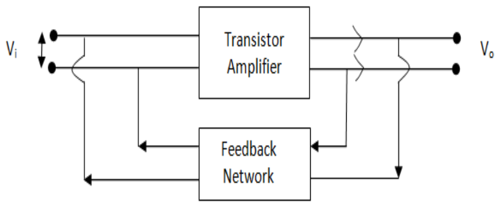
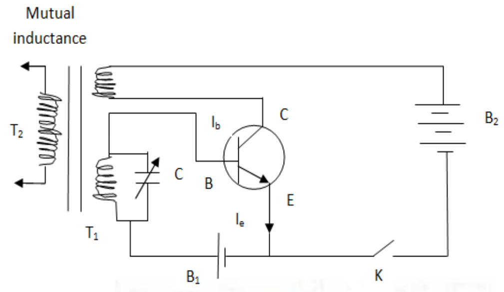


Working: When the reverse-bias voltage across p-n junction diode is increased, at a particular voltage the reverse current suddenly increases to a large value, called breakdown of diode, voltage is called breakdown voltage. In this voltage the rate of creation of hole-electron pairs increases thus current increases.

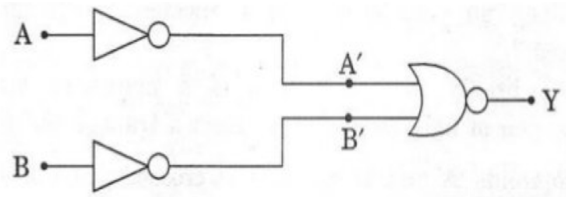
Zener diode is connected to unregulated DC voltage. Zener diode is operated in reverse bias. The current through R and Zener diode increases as voltage increased. The current through the diode increases by the voltage remains constant in breakdown region. I-V characteristics shown in graph shows voltage V_o across Load R_L is constant. If there is small change in input voltage V_i the current through RL remains same.

OR

(a) Explain briefly the principle on which a transistor amplifier works as an oscillator. Draw the necessary circuit diagram and explain its working.
Answer: Oscillator is a device which converts D.C power to alternating power of desired frequency. It is self-sustained transistor amplifier.

Block Diagram: 	As shown in the block diagram a portion of the output is fed back in phase to the input the process is called feedback. Amplification is done by (amplifier circuit) Common Emitter (CE) transistor. Feedback is done by Tank circuit.
Circuit Diagram: 	**Working**: The basic parts in the circuit are amplifier and LC network. The amplifier section is a transistor in common emitter mode and LC network is simply consists of an inductor and a capacitance. Change is current is due to change in magnetic field in inductor and electric field across capacitor. Batteries are used to bias the transistor. A part of input signal is returned back to the input section after passing through LC network. This signal is amplified is amplified by transistor and a part is again fed back to its input section. Thus it is self sustaining device. The component with the proper frequency ν_0 gets resonantly amplified and the output acts as a source of alternating voltage of that frequency. The frequency can be varied by varying L and C. Phase shift of 180^0 is made by CE amplifier and frequency of oscillation $$\nu_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}$$

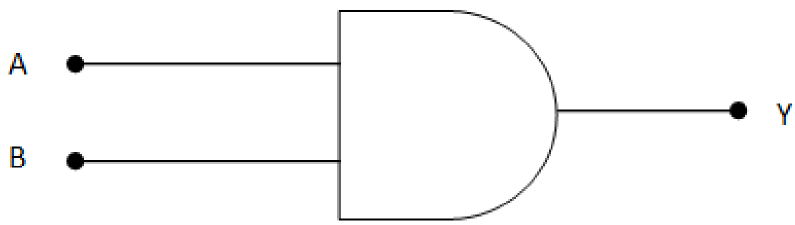
(b) Identify the equivalent gate for the following circuit and write its truth table.



Answer: Truth Table

A	B	A'	B'	Y=$\overline{A' + B'}$
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

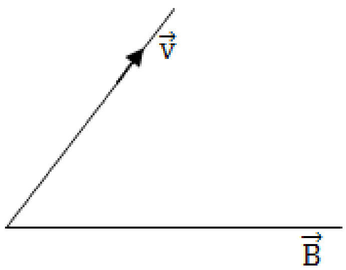
Thus from the truth table we find the input single A, B has output similar to AND operation.



Equivalent gate is AND as shown above.

Q30.(a) Write the expression for the force, $\vec{F}$, acting on a charged particle of charge 'q', moving with a velocity $\vec{v}$ in the presence of both electric field $\vec{E}$ and magnitude field $\vec{B}$. Obtain the condition under which the particle moves without deflection through the fields.

Answer: (a) Based on assumption that the direction of $\vec{E}$, $\vec{v}$ and $\vec{B}$ are such that $\vec{E} \perp \vec{v}$, $\vec{E} \perp \vec{B}$ and $\vec{v}$ is not parallel or anti parallel to $\vec{B}$.



Force due to electric field = $q\vec{E}$

Force due to magnetic field= $q(\vec{v} \times \vec{B})$

Therefore force due to electric and magnetic field

$$\vec{F} = q[\vec{E} + (\vec{v} \times \vec{B})]$$

For null deflection force due to electric field must balance the force due to magnetic field.

$$q\vec{E} = q(\vec{v} \times \vec{B})$$

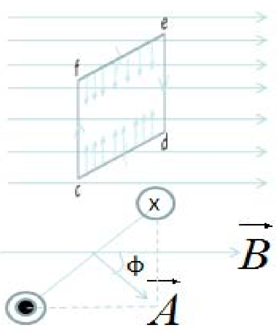
E=vBsin90^0

$$v = \frac{E}{B}$$

Velocity should be the ratio of magnitude of electric and magnetic field.

(b) A rectangular loop of size $l \times b$ carrying a steady current I is placed in a uniform magnetic field $\vec{B}$. Prove that the torque $\vec{\tau}$ acting on the loop is given by $\vec{\tau} = \vec{m} \times \vec{B}$, Where $\vec{m}$ is the magnetic moment of the loop.

Answer:



cdef is the rectangular coil of size $l \times b$

l & b = the length of each vertical & horizontal side of the coil respectively (assuming) the coil to be rectangular

i = the current flowing through the coil.

B = the induction vector of the magnetic field in which the coil is suspended.

We know that the force experienced by the current carrying conductor placed in a magnetic field is

$\vec{F} = i(\vec{l} \times \vec{B}) \rightarrow (1)$

Force on each wire 'de' and 'fc'

$|\vec{F}| = ibB\sin(90-\phi) = ibB\sin\phi$

where ϕ = angle between the normal to the plane of the coil and the direction of the magnetic field.

The direction of the force on the wires 'de' & 'fc' are in the plane of the coil along the downward and upward direction respectively and hence cancel out.

Force on each wire 'de' and 'fc'

$|\vec{F}| = ibB\sin(90-\phi) = ibB\sin\phi$

where ϕ = angle between the normal to the plane of the coil and the direction of the magnetic field.

The direction of the force on the wires 'de' & 'fc' are in the plane of the coil along the downward and upward direction respectively and hence cancel out.

Moment of the couple $\Gamma = ilB \times OC$

$\Gamma = ilBb\sin\phi = i(lb)B\sin\phi = iAB\sin\phi$

$$\tau = \vec{m} \times \vec{B}$$

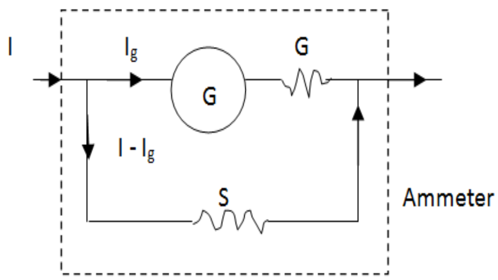
iA=$\vec{m}$ =magnetic moment.

OR

(a)Explain giving reasons, the basic difference in converting a galvanometer into (i) a voltmeter and (ii) an ammeter.

Answer: (i) A voltmeter measures potential drop. Ideally it should not draw any current. To convert a galvanometer to voltmeter a large resistance in series is connected to the Galvanometer.

(ii) An ammeter is used to measure current. The resistance should be zero. A galvanometer can be converted to ammeter by connecting a low resistance in parallel.



Let I_g be the current with which the galvanometer gives full scale deflection. Since galvanometer and shunt are connected in parallel then

P. D. Across galvanometer = P. D. Across shunt

$$\mathrm{Ig}\,G = (I - \mathrm{Ig})\,S$$

$$\therefore \quad S = \left(\frac{\mathrm{Ig}}{I - \mathrm{Ig}}\right) G$$

Hence by connecting a shunt of resistance S across the galvanometer, we get an ammeter of desired range.

Expression for the resistance of the ammeter:

Let R be the effective resistance of ammeter then

$$\frac{1}{R} = \frac{1}{G} + \frac{1}{S} = \frac{S+G}{GS}$$

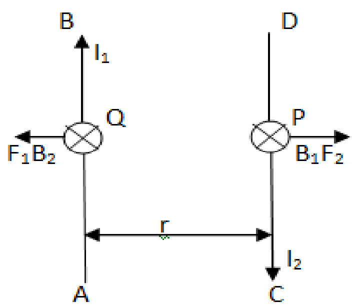
$$\text{Hence } R = \frac{GS}{G+S}$$

(a) Two long straight parallel conductors carrying steady currents I_1 and I_2 are separated by a distance 'd'. Explain briefly with the help of a suitable diagram how the magnetic field due to one conductor acts on the other. Hence deduce the expression for the force acting between the two conductors. Mention the nature of this force.

Answer: Consider two infinitely long thin conductors carrying currents in opposite directions.

Magnetic field B_1 due to I_1 at P on conductor CD is given by

$B_1 = \frac{\mu_0 I_1}{2\pi r}$



The magnetic field B_1 is perpendicular to plane of paper and directed inward. This field will produce a force/length F_2 on conductor given CD by

$F_2 = B_1 I_2 = \frac{\mu_0 I_1 I_2}{2\pi r}$ $[\because F = BIl, Here\ l = 1]$

By *Fleming's left hand rule* direction of F_2 is away from the conductor AB.

Similarly the current I_2 will create a field B_2 at Q directed inward which in turn will create force/length F_1

$\therefore F_1 = B_2 I_1 = \frac{\mu_0 I_1 I_2}{2\pi r}$

By *Fleming's left hand rule*, the direction of F_1 is away from the conductor AB. Hence the conductors repel each other.